

Datomic

Deep Dive



Who am I

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@mello at [#datomic](#) (clojurians slack)

don't panic,
it's a **safe space**

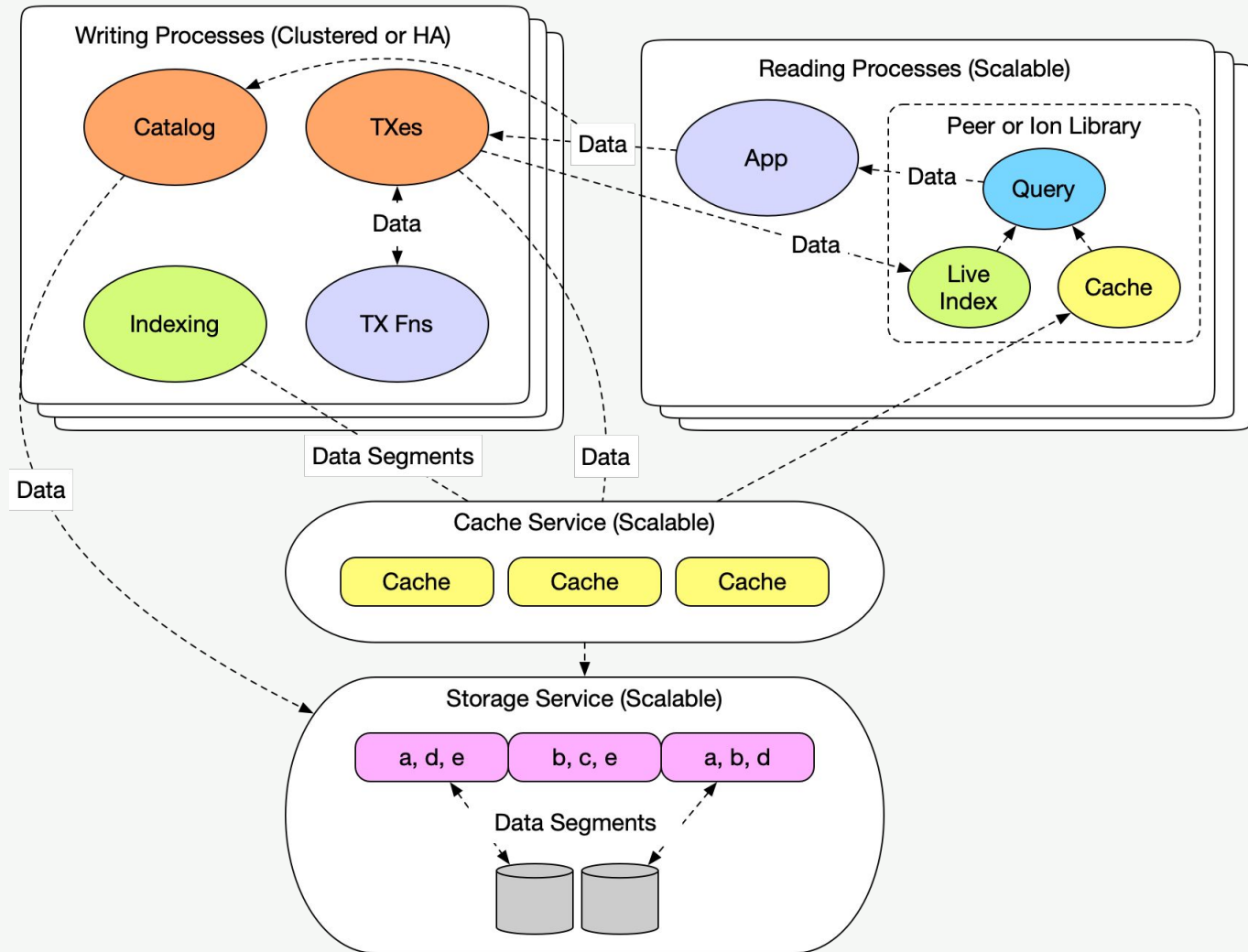
Agenda

- have fun
- take a closer look at transaction model
 - jepsten [report](#) made we re-review the documentation
- understand datomic-lab architecture
 - the repository shared earlier of the workshop
- import bank datasets
 - hands-on time in doubles (max threesome)
- you can interrupt *any time* if you have question or doubts
 - this workshop is for you!

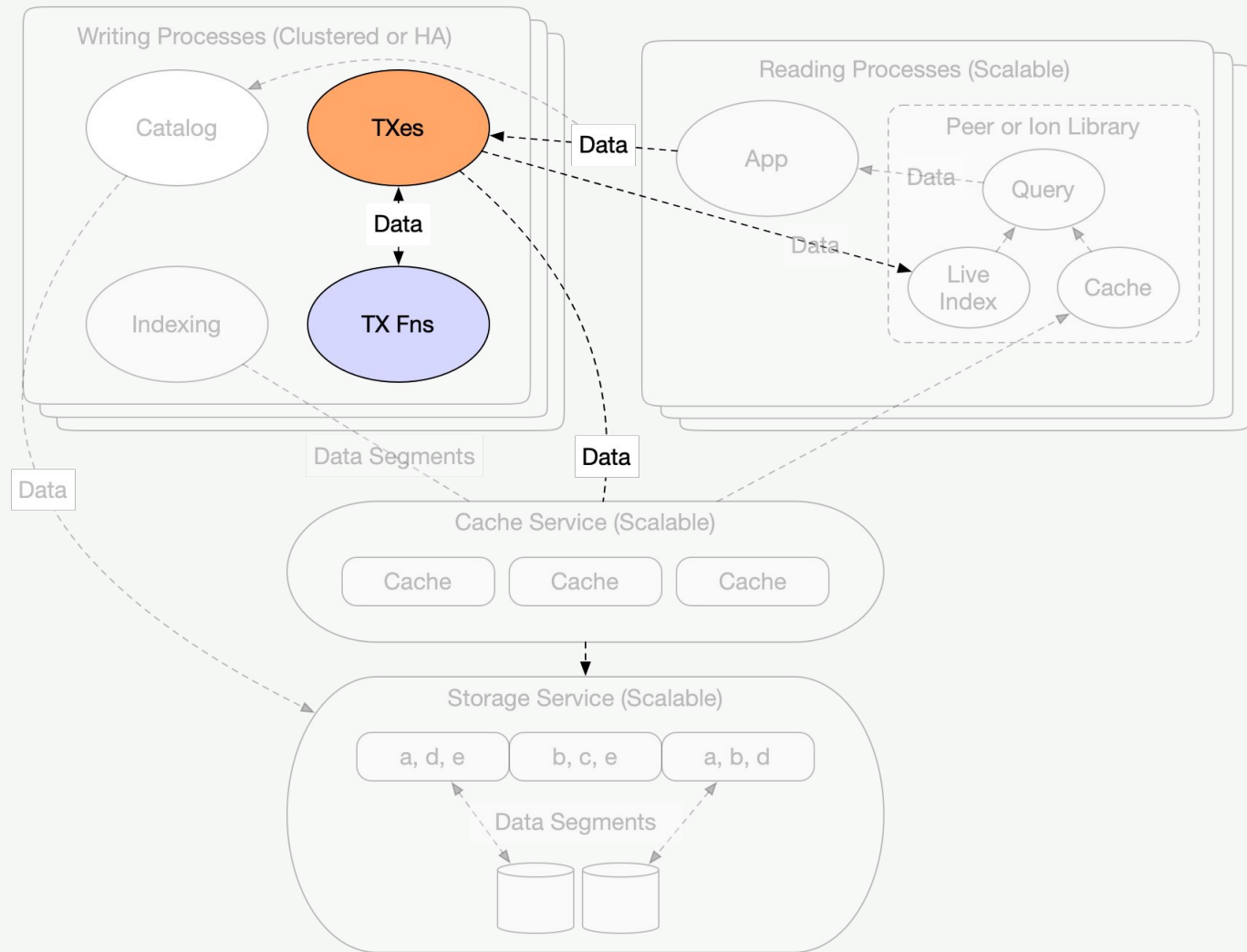
Definition of Success (DoS)

- your best friends from conj came from this workshop
- get confidence to use more Datomic

Datomic Architecture



Transactions



Transaction Model

- it's all about information
 - a Datomic database is *information*, specifically a set of atomic immutable facts. Because “fact” has many casual meanings, we call facts in Datomic *datoms*.
- d/with and d/transact
 - d/transact is nothing more then the composition of d/with with a durable swap operation
- application correctness
 - programs may wish to enforce business-rules validations at transaction boundaries.
 - programs may wish to generate, transform or validate input data with reference to data already in the database.

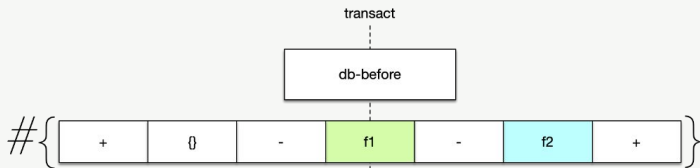
Transaction is a set of datoms,
not a sequence of commands.

*Just a reminder, it's all about **datoms***

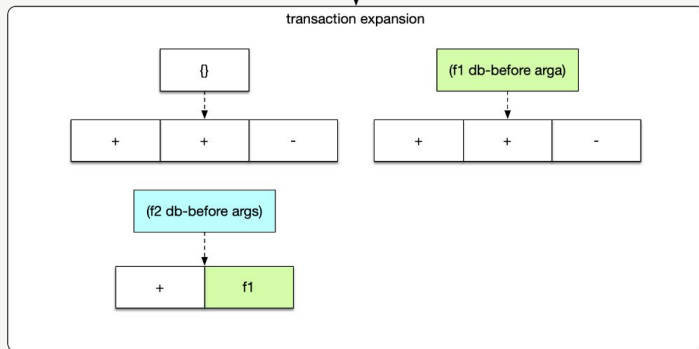
entity :e	attribute :a	value :v	transaction :tx	op :added
123	:person/name	"Mello"	455	true
123	:person/at	"Brazil"	455	true
123	:person/at	"Brazil"	456	false
123	:person/at	"USA"	456	true

Transaction Expansion

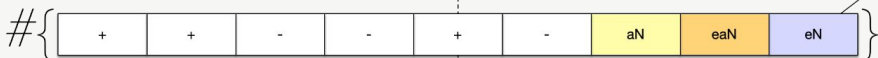
tx-data: mix of primitive list forms, map forms, and tx fn list forms



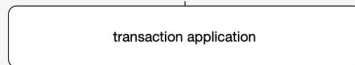
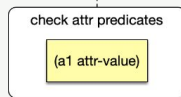
+	primitive assert list form
-	primitive retract list form
{}	map form
fN	transaction function
aN	attribute predicate
eaN	entity required attribute(s)
eN	entity predicate



after expansion:
primitive assert and retract list forms only

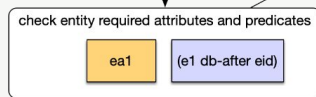
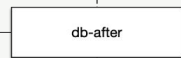


primitive asserts can be of db/ensure datoms, which trigger entity specs



predicates and specs compose as logical AND, all must be true

Datomic produces exactly one new database value per transaction

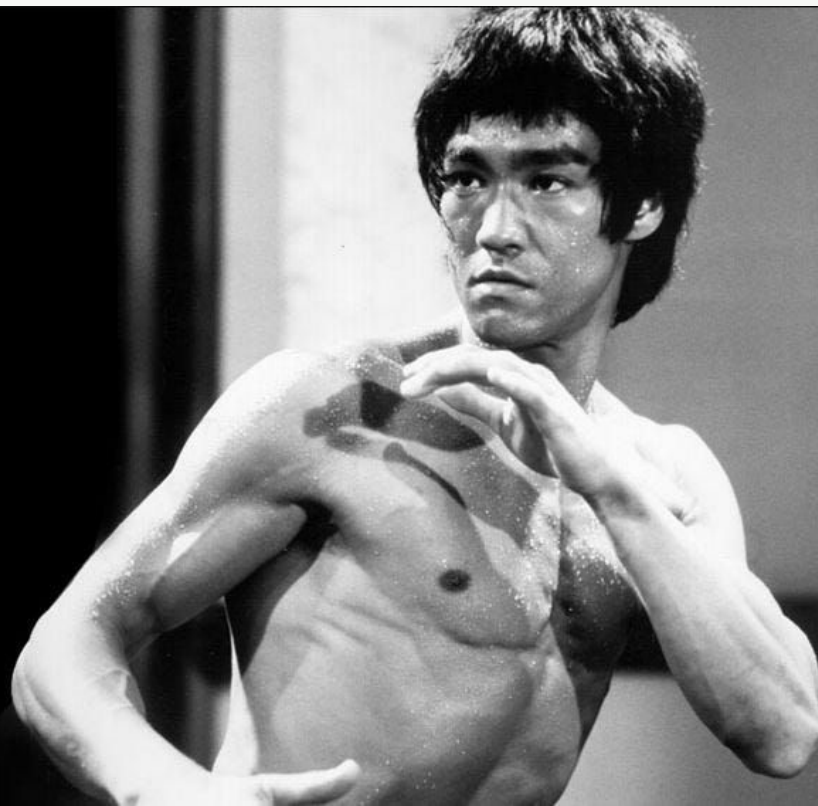


Application Correctness (*validations*)

- attribute spec (:db.attr/preds)
 - pure function
- entity spec required attributes (:db.entity/attrs)
- entity spec predicates (:db.entity/preds)
 - against **db-after**
- transaction functions
 - pure functions, never modifies the database
 - against db-before

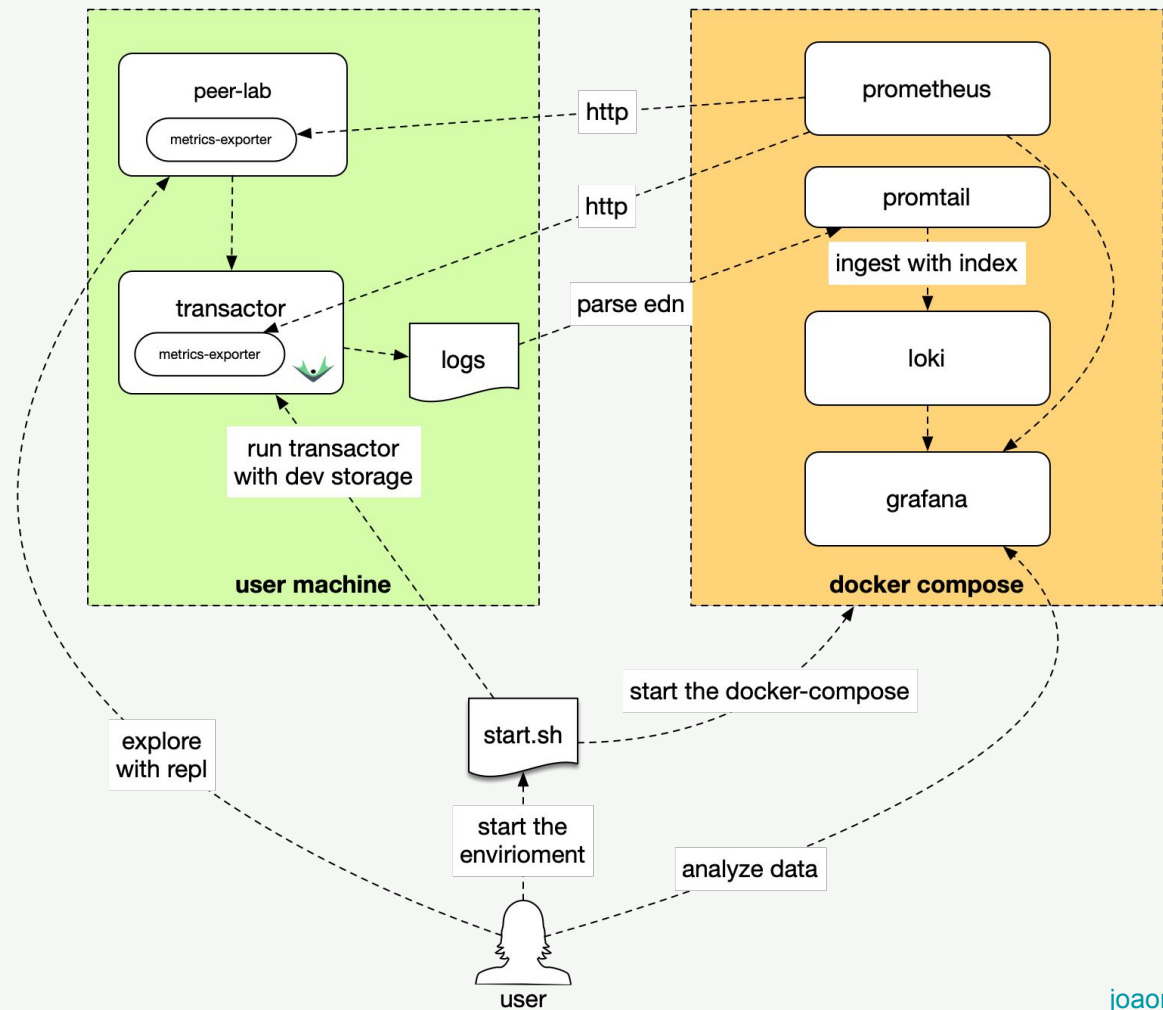
Knowing is not enough,
we must apply.
Willing is not enough,
we must do.

- Bruce Lee



(workshop :datomic-lab)

datomic-lab



It's REPL time

One database, many indexes

structure	attribute	contains
k/v	AVET	:db/unique or :db/index attributes
row	EAVT	all datoms
column	AEVT	all datoms
document	EAVT , partitions, components	all datoms
graph	VAET	:db.type/ref attribute